Assembly Instructions

Gas engine-generator sets

MTU 8V4000 GS – GG08V4000A1/A2 with engines 8V4000L32/L32FB/L33 and 8V4000L64
MTU 12V4000 GS – GG12V4000A1/A2 with engines 12V4000L32/L32FB/L33 and 12V4000L64
MTU 16V4000 GS – GG16V4000A1/A2 with engines 16V4000L32/L32FB/L33 and 16V4000L64
MTU 20V4000 GS – GG20V4000A1/A2 with engines 20V4000L32/L32FB/L33 and 20V4000L64

M060808/00E

Name	Model number	Engine model	Application group
MTU 8V4000 GS	GG08V4000A1	8V4000L32 8V4000L32FB 8V4000L33 8V4000L64	3A, continuous operation, unrestricted
	GG08V4000A2	8V4000L32 8V4000L32FB 8V4000L33 8V4000L64	3A, continuous operation, unrestricted
MTU 12V4000 GS	GG12V4000A1	12V4000L32 12V4000L32FB 12V4000L33 12V4000L64	3A, continuous operation, unrestricted
	GG12V4000A2	12V4000L32 12V4000L32FB 12V4000L33 12V4000L64	3A, continuous operation, unrestricted
MTU 16V4000 GS	GG16V4000A1	16V4000L32 16V4000L32FB 16V4000L33 16V4000L64	3A, continuous operation, unrestricted
	GG16V4000A2	16V4000L32 16V4000L32FB 16V4000L33 16V4000L64	3A, continuous operation, unrestricted
MTU 20V4000 GS	GG20V4000A1	20V4000L32 20V4000L32FB 20V4000L33 20V4000L64	3A, continuous operation, unrestricted
	GG20V4000A2	20V4000L32 20V4000L32FB 20V4000L33 20V4000L64	3A, continuous operation, unrestricted

Table 1: Applicability

Preface

This installation manual is intended to assist project planners, system/component designers and installation personnel in planning or carrying out the installation of MTU components.

The installation manual is applicable to all gas engines from the 4000 series (see business portal).

The objective of the installation manual is to facilitate the assembly of components.

The installation manual does not relieve those responsible for the system from independently ensuring that installation is carried out in an appropriate manner.

In addition to the assembly instructions, the following documents included in the delivery must be observed:
- Planning drawing
- Schematic diagrams
- TEN data
- Arrangement drawings
- Operating manual (see business portal)
- Maintenance schedule (see business portal)
- Fluids and lubricants specifications (see business portal)
- MTU standards

Each document is taken from the respective specifications document required for the order.

Adherence to the prescribed schedule for maintenance tasks will have a positive effect on operational safety, reliability and long service life. During the planning or installation of the plant system, ease of access for operation, maintenance and repair work is to be ensured.

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1 Safety

1.1 Important provisions for all products

General information

This product may pose a risk of injury or damage in the following cases:
- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications which are neither made nor authorized by the manufacturer
- Noncompliance with the safety instructions and warning notices

Nameplates

The product is identified by nameplate, model designation or serial number and must match with the information on the title page of this manual.

Nameplates, model designation or serial number can be found on the product.

All EU-certified engines delivered by MTU come with a second nameplate. When operating the machine in the EU: The second nameplate must be affixed in a prominent position as described in the accompanying specifications.

Emission regulations and emission labels

Responsibility for compliance with emission regulations

Modification or removal of any mechanical/electronic components or the installation of additional components including the execution of calibration processes that might affect the emission characteristics of the product are prohibited by emission regulations. Emission-related components must only be serviced, exchanged or repaired if the components used for this purpose are approved by the manufacturer.

Noncompliance with these specifications will invalidate the design type approval or certification issued by the emissions regulation authorities. The manufacturer does not accept any liability for violations of the emission regulations.

The product must be operated over its entire life cycle according to the conditions defined as "Intended use" (→ Page 9).

Emission certification applicable to engines with EPA Nonroad Tier 4 emission certification in accordance with 40 CFR 1039

Extract from the standard:

Failing to follow these instructions when installing a certified engine in a piece of nonroad equipment violates federal law (40 CFR 1068.105(b)), subject to fines or other penalties as described in the Clean Air Act.

Extract from the standard:

If you install the engine in a way that makes the engine's emission control information label hard to read during normal engine maintenance, you must place a duplicate label on the equipment, as described in 40 CFR 1068.105.

When fitting the second label, the requirements of 40 CFR 1068.105(c) must be followed and observed. This paragraph described the process for requesting and fitting the label, the documentation obligations and storage obligations for the required documents.

Replacing components with emission labels

On all MTU engines fitted with emission labels, these labels must remain on the engine throughout its operational life.

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Exception: Engines used exclusively in land-based, military applications other than by US government agencies.

Please note the following when replacing components with emission labels:
- The relevant emission labels must be affixed to the spare part.
- Emission labels shall not be transferred from the replaced part to the spare part.
- The emission labels must be removed from the replaced part and destroyed.

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1.2 Intended use of MTU Onsite Energy GmbH gas systems

Intended use

The machine is intended to be used exclusively for electrical power generation and waste heat utilization in stationary operation, either in sufficiently ventilated rooms or in container solutions.

A gas/air mixture is fed to the reciprocating internal combustion engine. This is converted into mechanical work in the individual cylinders through combustion. The flywheel of the engine drives the synchronous generator connected via an elastic coupling, resulting in a conversion to electric energy. The engine and the generator are joined to the base frame with elastic, vibration-damped elements. For a mains frequency of 60 Hz, a gearbox is placed between the engine and the generator. The waste heat generated during the combustion process is dissipated to the surroundings via coolers (power generator) and/or made available to the customer via heat exchangers (combined heat and power plant).

Area of application	Suitable	Comments
Industry	Yes	–
Business/Service	Yes	In some individual cases, however, such as with swimming pool facilities, schools, heating stations for apartment buildings and the like, there is no end user in the classical sense involved.
Household	No	Operation is excluded
Potentially explosive area	No	Operation is not permitted in potentially explosive areas
Underground	No	Operation is excluded
Mobile use	No	Operation is excluded

Table 2: Area of application of machine

There are 3 operating modes of the machine:
- Mains parallel operation
- Mains failure operation
- Isolated operation

Foreseeable misuse

Any use other than that specified under "Intended use" or any extended use, in particular mishandling, is regarded as improper use. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper use.

Operation

Operation is not permissible if the physical conditions specified on the technical data sheet or on/in other documents (concerning temperature, pressure, site altitude, etc.) are not complied with. Operation is not permissible without protective equipment, usage of approved spare parts, exclusion of conversions, warning of foreseeable misuse.

Operation is not permissible without the use of fluids and lubricants from MTU Onsite Energy (natural gas, biogas, quality-compliant gas, quality-compliant water, lubricants) in accordance with the (→ fluids and lubricants specifications).

Operation is not permissible without adopting the preservation measures approved by MTU Onsite Energy according to the (→ preservation and represervation specifications).

The specifications of the manufacturer will be amended or supplemented as necessary. Prior to operation, make sure that the latest version of these are adopted. The latest version can be found at:
- http://www.mtuonsiteenergy.com

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Surroundings, climate, temperature

The weather plays no role with regard to the intended use of the genset/module provided that the operation takes place inside an installation room. For container applications, however, the permissible outdoor temperatures at the installation site plays a decisive role. The product-specific ambient conditions such as climate and temperature are indicated in the technical description.

Prerequisite for proper use

Fulfillment of all the requirements set out in the “Installation conditions” document.

Any use that does not fall within the described limits for use is regarded as improper.

Modifications or conversions

Unauthorized changes to the product breach the conditions concerning its intended use and compromise safety.

Only modifications or conversions expressly authorized by the manufacturer are regarded as compliant in this sense. The manufacturer shall not be held liable for any damage resulting from unauthorized modifications or conversions.

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